WZORY TRYGONOMETRYCZNE

$\operatorname{tg} x = \frac{\sin x}{\cos x}$
$\operatorname{ctg} x = \frac{\cos x}{\sin x}$
$\sin 2x = 2 \sin x \cos x$
$\cos 2x = \cos^2 x - \sin^2 x$

$\sin^2 x = \frac{1-\cos 2x}{2}$
$\cos^2 x = \frac{1+\cos 2x}{2}$
$\sin^2 x + \cos^2 x = 1$

ASYMPTOTY UKOŚNE

$$y = mx + n$$

$m = \lim\limits_{x \to \pm\infty} \frac{f(x)}{x}$, $n = \lim\limits_{x \to \pm\infty} [f(x) - mx]$

POCHODNE

$[f(x) + g(x)]' = f'(x) + g'(x)$
$[f(x) - g(x)]' = f'(x) - g'(x)$
$[cf(x)]' = cf'(x)$, gdzie $c \in \mathbb{R}$
$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$
$\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}$, o ile $g(x) \neq 0$
$[f\,(g(x))]' = f'(g(x))g'(x)$
$[f(x)]^{g(x)} = e^{g(x) \ln f(x)}$
$(c)' = 0$, gdzie $c \in \mathbb{R}$
$(x^p)' = px^{p-1}$
$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$
$(\frac{1}{x})' = -\frac{1}{x^2}$
$(a^x)' = a^x \ln a$
$(e^x)' = e^x$
$(\log_a x)' = \frac{1}{x \ln a}$
$(\ln x)' = \frac{1}{x}$
$(\sin x)' = \cos x$
$(\cos x)' = -\sin x$
$(\operatorname{tg} x)' = \frac{1}{\cos^2 x}$
$(\operatorname{ctg} x)' = -\frac{1}{\sin^2 x}$
$(\operatorname{arc\,sin} x)' = \frac{1}{\sqrt{1-x^2}}$
$(\operatorname{arc\,cos} x)' = -\frac{1}{\sqrt{1-x^2}}$
$(\operatorname{arc\,tg} x)' = \frac{1}{1+x^2}$
$(\operatorname{arc\,ctg} x)' = -\frac{1}{1+x^2}$